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Inventor(s)

Leung; Kai Yip

SYSTEM AND METHOD FOR ESTABLISHING CALLS

Abstract

An example method includes: receiving, from an initiating device, a call destined for a target account; establishing, at a server, a central call session for the call and connecting the initiating device to the central call session; sending a central call session notification to a target device associated with the target account, the central call session notification to prompt the target device to initiate a request to join the central call session; receiving, from the target device, a response to the central call session notification; and when the response comprises the request to join the central call session, connecting the target device to the central call session to establish the call.

Inventors: Leung; Kai Yip (Fremont, CA)

Applicant: TEXTNOW, INC. (Waterloo, CA)

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Background/Summary

FIELD

[0001] The specification relates generally to telecommunication services, and more particularly to a

system and method for establishing calls.

BACKGROUND

[0002] Telecommunications services use an exchange of messages to provision the correct information to establish a communications session. In particular, signaling protocol specifies the registration of devices prior to connecting an incoming call. However, in modern systems, many devices are already registered with their respective servicing servers, resulting in redundant and time-consuming exchanges.

SUMMARY

[0003] According to an aspect of the present specification an example method includes: receiving, from an initiating device, a call destined for a target account; establishing, at a server, a central call session for the call and connecting the initiating device to the central call session; sending a central call session notification to a target device associated with the target account, the central call session notification to prompt the target device to initiate a request to join the central call session; receiving, from the target device, a response to the central call session notification; and when the response comprises the request to join the central call session, connecting the target device to the central call session to establish the call.

[0004] According to another aspect of the present specification, an example method includes: receiving, at a target device, a central call session notification from a server, the central call session notification indicating a call; signaling the central call session notification to prompt a response to the central call session notification; and when the response comprises an acceptance of the call, sending a request to join the central call session to the server to accept the call.

[0005] According to another aspect of the present specification, an example server includes: a memory and a communications interface; a processor interconnected with the memory and the communications interface, the processor configured to: receive, from an initiating device, a call destined for a target account; establish a central call session for the call and connecting the initiating device to the central call session; send a central call session notification to a target device associated with the target account, the central call session notification to prompt the target device to initiate a request to join the central call session; receive, from the target device, a response to the central call session notification; and when the response comprises the request to join the central call session, connect the target device to the central call session to establish the call.

[0006] According to another aspect of the present specification, an example computing device comprising: a memory and a communications interface; and a processor interconnected with the memory and the communications interface, the processor configured to: receive, at a target device, a central call session notification from a server, the central call session notification indicating a call; signal the central call session notification to prompt a response to the central call session notification; and when the response comprises an acceptance of the call, send a request to join the central call session to the server to accept the call.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] Implementations are described with reference to the following figures, in which:

[0008] FIG. 1 depicts a schematic diagram of an example system for establishing calls using central call sessions.

[0009] FIG. 2 depicts a block diagram of certain internal components of the server and the device of FIG. 1.

[0010] FIG. 3 depicts a flowchart of an example method for establishing calls using central call sessions.

[0011] FIG. 4 depicts a schematic diagram of an example performance of establishing a call in the

system of FIG. 1.

[0012] FIG. 5 depicts a flowchart of an example method of reconnecting to a call using central call sessions.

[0013] FIG. 6 depicts a flowchart of an example method of merging calls using central call sessions.

DETAILED DESCRIPTION

[0014] Call services may establish calls using a series of industry-standardized communications protocols, such as the session initiation protocol (SIP) to initialize communications sessions between endpoint devices (i.e., a source device from which the communication session is initiated, and a target device for which the communication session is destined).

[0015] In particular, as part of the standardized SIP exchange for an incoming call destined for a target device, the target device is configured to register with the servicing server. This registration process requires extra exchanges and hence takes extra time and can cause delays and/or failures in establishing the call.

[0016] Accordingly, as described herein, an example system for establishing calls using call sessions is provided. In particular, the system, and more specifically, the call service server, receives an incoming call destined for a target account and, rather than pushing a call notification to the target device, thereby initiating the registration process, the server establishes a call session hosted centrally at the server (i.e., a central call session). The server may then send a central call session notification to the target device to prompt the target device to initiate a request to join the central call session. In particular, the target device may display the central call session notification to an operator or user as an incoming call, and may process the central call session notification by responding with the request to join the central call session or a rejection of the call. Accordingly, from the perspective of the target device, the request to join the central call session may be processed as an outgoing call (i.e., to the central call session), thereby simplifying the signaling process. The server may then process the request to join the central call session by connecting the target device to the central call session, thereby effectively establishing the call between the initiating device and the target device via the central call session.

[0017] FIG. 1 depicts a system **100** for establishing calls using call sessions. The system **100** includes a call service server **104** (also referred to herein as simply the server **104**), computing devices **108**, **110** interconnected with the server **104**, a network **112**, and a second computing device **116**.

[0018] The server **104** is generally configured to provide calling services to support calls between computing devices, such as the computing devices **108**, **110** (also referred to herein as simply the device **108** and the device **110**) and the second computing device **116**. The server **104** may be any suitable server environment, including a series of cooperating servers, one or more cloud-based servers, and the like. The internal components of the server **104** will be described in greater detail below.

[0019] The computing devices **108**, **110** are generally configured to support calling applications, to allow a user of the computing devices **108**, **110** to engage in calls (e.g., voice calls, video calls, or the like) with a user of another computing device, such as the second computing device **116**. The computing devices **108**, **110** may be computing devices such as mobile phones, tablets, laptop computers, desktop computers, or the like. The internal components of the computing device **108** will be described in greater detail below.

[0020] The server **104** is in communication with the computing device **108** via one or more communication links, shown in the present example as being wireless communication links. The communication links may be wired, wireless, a combination thereof, and may traverse one or more networks, including local area networks, wide area networks, the internet, and the like. In particular, in the present example, call services and call handling for the computing devices **108**, **110** is provided by the call service server **104**. That is, the user of the computing devices **108**, **110**

may subscribe to the call service provided by the server **104**. The server **104** may therefore, for example, provision a user account for the user, a telephone number associated with the user account, and the like.

[0021] The server **104** is further in communication with the network **112**. The network **112** may include any number of networks, including mobile networks, wide area networks, local area networks, public switched telephone networks (PSTNs), combinations of the above, and the like.

[0022] The system **100** further includes the second computing device **116**, which may be a mobile phone, a tablet, a laptop computer, a desktop computer, or the like. In the present example, the second computing device **116** may be serviced by a call service provider other than the server **104**, and which may communicate with the server **104** via the network **112**.

[0023] The system **100**, and in particular, the server **104** is configured to establish calls using call sessions, or central call sessions, hosted at the server **104**. In particular, upon receiving an incoming call from the second computing device **116** (i.e., also referred to herein the initiating device **116**), the server **104** establishes a central call session at the server **104** and adds or connects the initiating device **116** to the central call session. The server **104** may then send an invitation to one or both of the target devices **108**, **110** (i.e., as being associated with the target account to which the incoming call is directed) to prompt the target device **108**, **110** to join the central call session. Thus, at the target device **108**, **110**, the call may be structured as an outgoing call to the central call session, rather than an incoming call from the initiating device **116**. Accordingly, the call may be more robust and protected against lost calls due to network or other connectivity issues, since the outgoing call to the central call session may be re-established by either party.

[0024] Turning now to FIG. 2, certain internal components of the server **104** and the computing device **108** are depicted in greater detail. The server **104** includes a processor **200**, a memory **204** and a communications interface **208**.

[0025] The processor **200** may include a central processing unit (CPU), a microcontroller, a microprocessor, a processing core, a field-programmable gate array (FPGA), or similar. The processor **200** may include multiple cooperating processors. The processor **200** may cooperate with the memory **204** to realize the functionality described herein.

[0026] The memory **204** may include a combination of volatile (e.g., Random Access Memory or RAM) and non-volatile memory (e.g., read-only memory or ROM, Electrically Erasable Programmable Read Only Memory or EEPROM, flash memory). All or some of the memory **204** may be integrated with the processor **200**. The memory stores applications, each including a plurality of computer-readable instructions executable by the processor **200**. The execution of the instructions by the processor **200** configures the server **104** to perform the actions discussed herein. In particular, the applications stored in the memory **204** include a server-side call handling application **212**. When executed by the processor **200**, the application **212** configures the processor **200** to perform various functions discussed below in greater detail and related to the call handling operation of the server **104**. The application **212** may also be implemented as a suite of distinct applications. Further, Some or all of the functionality of the application **212** may be implemented as dedicated hardware components, such as one or more FPGAs or application-specific integrated circuits (ASICs).

[0027] The memory **204** also stores rules and data for the call handling operation. For example, the memory **204** may include a user account repository **216** storing user account information such as user and/or account identifiers, telephone numbers, mobile directory numbers, or the like associated with the user account, contact and billing information for the user account, computing devices associated with the user account, and the like. For example, in some examples, the devices **108**, **110** may be separate computing devices associated with the same user account, while in other examples, each of the devices **108**, **110** may be associated with different user accounts.

[0028] The memory **204** may further include a call session tracking repository **220** storing call session data, including call session identifiers for ongoing call sessions managed by the server **104**,

optionally, status identifiers for each of the call identifiers, and the like. For example, some status identifiers may include 'pending' to indicate that the corresponding call has not yet been answered by the target account, 'cancelled' to indicate that the corresponding call was cancelled (e.g., by the source device), 'rejected' to indicate that the target account has rejected the call, and the like. In some examples, further device and/or account information may also be tracked in the call session tracking repository **220**. For example, the call session tracking repository **220** may include identifiers (e.g., account identifiers, Internet Protocol (IP) addresses, telephone numbers, etc.) for the participants of each call session, as well as a connectivity status for each participant.

[0029] The server **104** further includes the communications interface **208** interconnected with the processor **200**. The communications interface **208** may be configured for wireless (e.g., satellite, radio frequency, Bluetooth, Wi-Fi, or other suitable communications protocols) or wired communications and may include suitable hardware (e.g., transmitters, receivers, network interface controllers, and the like) to allow the server **104** to communicate with other computing devices. The specific components of the communications interface **208** are selected based on the types of communication links that the server **104** communicates over.

[0030] In some examples, the server **104** may further include one or more input and/or output devices (not shown). The input devices may include one or more buttons, keypads, touch-sensitive display screen, mice, or the like for receiving input from an operator. The output devices may include one or more display screens, monitors, speakers, sound generators, vibrators, or the like for providing output or feedback to an operator.

[0031] The device **108** includes a processor **230**, a memory **234**, a communications interface **238**, and one or more input and/or output devices **242**.

[0032] The processor **230** is similar to the processor **200** and may include a CPU, a microcontroller, a microprocessor, a processing core, an FPGA, or similar. The processor **230** may include multiple cooperating processors. The processor **230** may cooperate with the memory **234** to realize the functionality described herein.

[0033] The memory **234** is similar to the memory **204** and may include a combination of volatile and non-volatile memory, and all or some of the memory **234** may be integrated with the processor **230**. The memory **234** stores applications, including a device-side call handling application **246**. The application **246** includes a plurality of computer-readable instructions executable by the processor **230**. The execution of the instructions by the processor **230** configures the device **108** to perform the actions discussed herein. The application **246** may be implemented as a suite of distinct applications. Further, some or all of the functionality of the application **246** may be implemented as dedicated hardware components.

[0034] The memory **234** also stores rules and data for the call handling operation. For example, the memory **234** may include one or more repositories, to track call logs, call session identifiers, and the like.

[0035] The device **108** further includes the communications interface **238** interconnected with the processor **230**. The communications interface **238** may be configured for wireless or wired communications and may include suitable hardware to allow the device **108** to communicate with other computing devices, such as the server **104**. The specific components of the communications interface **238** are selected based on the types of communication links that the device **108** communicates over.

[0036] The device **108** further includes the one or more input and/or output devices **242**. The input devices may include one or more buttons, keypads, touch-sensitive display screen, mice, or the like for receiving input from an operator. The output devices may include one or more display screens, monitors, speakers, ringers, sound generators, vibrators, or the like for providing output or feedback to an operator.

[0037] FIG. **3** illustrates a method **300** of handling calls using central call sessions. The method **300** will be discussed in conjunction with its performance in the system **100**, and particularly by the

server **104**, via execution of the application **212**, and the device **108**, via execution of the application **246**. In particular, the method **300** will be described with reference to the components of FIGS. **1** and **2**. In other examples, the method **300** may be performed by other suitable devices or systems.

[0038] At block **305**, the server **104** receives an incoming call destined for a target account. For example, the target of an incoming call may be specified by a target telephone number. In particular, the incoming call may be received, in the present example, from the second computing device **116** via the network **112**. That is, the second computing device **116** may act as an initiating device from which the incoming call originates. More particularly, the server **104** may receive the incoming call from a PSTN, another server implementing a call service for the initiating device **116**, combinations of the above, or the like.

[0039] In some examples, the server **104** may receive the incoming call as a session initiation protocol (SIP) message, and more particularly, an [INVITE] message. In other examples, other suitable signaling protocols may be employed. The incoming call may have a call identifier associated with it, for example embedded in the SIP [INVITE] message or other signaling protocol message.

[0040] At block **310**, in response to receiving the incoming call, the server **104** processes the incoming call. In particular, the server **104** establishes a central call session hosted at the server **104** for the incoming call. The central call session may be a communications session accessible by multiple parties and/or devices, similar to a conferencing session. In contrast to conferencing sessions, the central call session may be a temporary session hosted and owned by the server **104** and dissociated from the target account to which the incoming call is targeted. That is, a conferencing session may be owned by a specific target account which is targeted by an incoming call to join the conferencing session, and accordingly is persistent in existence and in accessibility by targeting the target account. The central call session is established in response to an incoming call for a target account, but is not owned by the target account, and hence is transient and may not be accessible at the termination of the central call session.

[0041] The central call session may be identified by a central call session identifier and recorded in the call session tracking repository **220**. That is, the server **104** may be configured to create an entry in the call tracking repository **220** including the call session identifier for the central call session established for the incoming call. For example, the call session identifier may be stored in association with the call identifier for the incoming call and an identifier of the initiating device **116**. The server **104** may then add or connect the initiating device **116** to the central call session. In some examples, the server **104** may additionally provide a pending indication at the central call session that the call received from the initiating device **116** is pending, for example by providing an auditory ringing tone or indication at the central call session.

[0042] Additionally, at block **310**, the server **104** is configured to extract the target account and identify one or more devices associated with the target account. For example, the server **104** may retrieve, from the account repository **216**, a set of devices (e.g., by IP address or similar) associated with the target account and/or the target telephone number. For example, the server **104** may identify the device **108** as a target device, in the case that the devices **108**, **110** are associated with separate accounts. In other examples, if both the devices **108**, **110** are associated with the same target user account, then both of the devices **108**, **110** may be identified in the set of target devices. Thus, for example, if a user has a cell phone, a desktop computer, and a tablet, each of which is logged in with the same user account, then each of the devices may be identified as a target device in the set.

[0043] At block **315**, the server **104** sends a central call session notification to the target device identified at block **310**. In examples where a set of multiple target devices is identified at block **310** as being associated with the target account, the server **104** may send a central call session notification to each target device in the set, or a default target device selected from the set. In the

present example, the central call session notification is sent from the server **104** to the device **108**. [0044] In contrast to conventional call establishment flows, the central call session notification sent at block **315** may be distinct and unrelated to a conventional call notification which signals an incoming call to the target device **108** to prompt the target device **108** to register with the server **104** (e.g., via a SIP [REGISTER] action), thereby allowing the server **104** to forward the SIP [INVITE] message to the target device **108**.

[0045] Rather, the central call session notification may signal the central call session and the initiating device **116** to the target device **108** to prompt the target device **108** to either reject or join the central call session, as will be further described herein. Notably, the central call session notification does not prompt the target device **108** to register with the server **104**.

[0046] The central call session notification may be sent, for example using Firebase Cloud Messaging (FCM), Apple Push Notification service (APNs), or other suitable notification services which allow notifications to be sent to the target device. In particular, the central call session notification may include an indication of the call session, the call session identifier, and an indicator of the initiating device **116** (e.g., a telephone number).

[0047] At block **320**, the device **108** receives the central call session notification from the server **104**. In particular, the central call session notification includes the indication of the call session, the call session identifier, and the indicator of the initiating device **116** (e.g., a telephone number). The device **108** may store the extracted data, and in particular, the central call session identifier, for example in the memory **234**, such as in an application cache or the like, as a pending call.

[0048] In response to receiving the central call session notification, the device **108** may signal the central call session to the user. For example, the device **108** may generate an alert, such as an auditory or mechanical (e.g., vibrational) ring for example based on volume settings at the device **108**, a visual notification (e.g., a pop-up), or the like. More particularly, the alert may be provided using one or more of the output devices of the device **108**. In particular, the alert may be presented to the user similarly to a call alert signaling an incoming call from the initiating device **116**, for example by identifying the initiating device by telephone number, or other indicator, such as a contact name stored at the device **108** or in association with the user account at the server **104**. Accordingly, the alert may similarly provide options for accepting or rejecting the call from the initiating device **116**.

[0049] In some examples, in response to receiving the central call session notification, prior to signaling the central call session to the user, the device **108** may initialize a connection with the server **104** in preparation for later signaling (e.g., to signal the acceptance or rejection of the call). For example, the device **108** may send an initial SIP [INVITE] message to initialize the connection. In some examples, signaling the central call session to the user and initializing the connection may occur substantially simultaneously.

[0050] At block **325**, the device **108** determines whether the call from the initiating device **116** is accepted (answered) or rejected (not answered). For example, the device **108** may make the determination based on user input to the alert or expiry of a timeout of the alert. For example, if the user indicates acceptance of the call, the device **108** may make an affirmative determination at block **325** to respond affirmatively to the central call session notification. If the user indicates rejection of the call, the device **108** may make a negative determination at block **325** to respond negatively to the central call session notification. If the alert times out (e.g., a predefined amount of time period expires), then the invitation may timeout and the device **108** may similarly make a negative determination at block **325** to respond negatively to the central call session notification.

[0051] If, at block **325**, the determination is affirmative, then the device **108** proceeds to block **330**. At block **330**, the device **108** sends a response to the central call session notification to the server **104**. In particular, the response may include a request to join the central call session. For example, the request may be a SIP [INVITE] message, or other standard invitation or call initiation according to other signal protocols. Further, the device **108** may include the call session identifier

in the SIP [INVITE] message to indicate the central call session which the device **108** is requesting to join. In particular, the SIP [INVITE] message may be a secondary or subsequent SIP [INVITE] message sent via the initially established connection between the device **108** and the server **104**. [0052] At block **335**, the server **104** receives the request to join the central call session from the device **108** and in response, adds or connects the device **108** to the central call session identified in the request. In particular, having connected both the initiating device **116** and the target device **108** to the central call session, the initiating device **116** and the target device **108** may exchange media and communications, thereby effectively establishing the incoming call from the initiating device **116**. Accordingly, the server **104** may additionally terminate a ringing indication at the central call session.

[0053] If, at block **325**, the determination is negative, then the device **108** proceeds to block **340**. At block **340**, the device **108** sends a response to the call session notification to the server **104**. In particular, the response may include a rejection of the central call session. The rejection may be sent via the initially established connection. For example, the rejection may be a different type of SIP request, such as a [CANCEL] action (e.g., in the case that a SIP [200OK] message is not received in response to the initial SIP [INVITE]), a [BYE] action, or similar, to indicate to the server **104** that the device **108** is rejecting the incoming call from the initiating device **116**. The device **108** may therefore include the call session identifier to indicate the central call session which the device **108** is rejecting. The device **108** may further include a custom header in a SIP [BYE] message, for example to indicate whether the rejection is caused by a timeout or an active rejection by a user.

[0054] At block **345**, the server **104** receives the rejection of the central call session from the device **108** and in response, terminates the central call session identified in the rejection. In such examples, the server **104** may provide an indication of the rejection at the central call session to indicate to the initiating device **116** that the call was rejected. In some examples, when the rejection is caused by a timeout, the server **104** may redirect the initiating device **116** to a voicemail for the device **108**.

[0055] FIG. **4** depicts a schematic diagram of a call handling operation **400** in the system **100** for a successful connected call, in accordance with the present disclosure. The operation **400** will be described in conjunction with performance of the method **300**, where reference numerals of the operation **400** correspond to blocks of the method **300**. Further, the operation **400** will be described in context of SIP signaling. In other examples, other suitable methods and signaling protocols may be employed to achieve the call handling operation **400**.

[0056] The operation **400** is initiated when the device **116** sends a call invitation **405**, in the form of an [INVITE] message having a call identifier, in the present example, call-id:123. The call invitation **405** may further include source and target parameters (e.g., 'to' and 'from' fields) and the like (not shown).

[0057] The call invitation **405** may be received by the server **104** via the network **112** at block **305**. The server **104** may then establish a central call session **410**, having a central call session identifier, ccs-id:456. The server **104** may additionally connect the device **116** to the central call session **410** having identifier ccs-id:456.

[0058] The server **104** additionally sends a central call session notification (CCS not.) **415**. The central call session notification **415** may be a push notification or similar by a third-party service. In particular, the central call session notification **415** includes the central call session identifier, ccs-id:456.

[0059] In response to the central call session notification, the device **108** generates an alert **420** (i.e., at block **320**) to prompt a response to the server **104**. The response may be a rejection (not shown) or an acceptance. In the case of an acceptance or affirmative response, the device **108** initiates a SIP [INVITE] request **430** (i.e., at block **330**) to the server **104** indicating the central call session identifier, ccs-id:456.

[0060] In response to the SIP [INVITE] request **430**, the server **104** connects the device **108** to the

central call session **410** having central call session identifier ccs-id:456. Thus, the server **104** establishes the call **435** between the initiating device **116** and the target device **108** via the central call session **410** having central call session identifier ccs-id:456.

[0061] Calls established using call sessions hosted centrally at the server **104** may therefore be established more quickly and efficiently and more reliably, since there are fewer exchanges between the device **108** and the server **104**, for example to perform a SIP [REGISTER] process prior to sending the SIP [INVITE] request.

[0062] Upon establishing the central call session, either endpoint device **108** or **116** may end the call, for example by indicating a hang-up action on the respective device. The hang-up action may send a SIP [BYE] message to the server **104** and the central call session **410**, which may forward a SIP [BYE] message to the other endpoint device **108** or **116** to end the call **435**.

[0063] Further, since the device **108** initiates the SIP [INVITE] request, the subsequently established call (i.e., via the central call session) may be better protected against temporary network connectivity issues or the like. For example, referring to FIG. 5, an example method **500** of reconnecting to a central call session is depicted.

[0064] At block **505**, the device **108** is in an ongoing call, via a central call session at the server **104**, with the device **116**. In some examples, the device **108** may be the target device as in the examples described above, while in other examples, the device **108** may be the initiating device of the ongoing call. That is, the method **500** may be applied by the server **104** and disconnecting device **108** irrespective of the origination of the ongoing call.

[0065] At block **510**, the device **108** detects a loss of connectivity to the central call session. For example, the device **108** may be travelling through a region of poor network service, the device **108** may be experiencing difficulty or delays roaming from one access point to another, or similar. In particular, the loss of connectivity to the central call session may be an unexpected loss of connectivity, and not precipitated by an active termination by the device **108** or the device **116** (e.g., via the central call session).

[0066] Similarly, at block **515**, the server **104** may detect the loss of connectivity of the device **108** from the central call session. The server **104** may provide an indication at the central call session that the device **108** has lost connectivity. In some examples, the server **104** may facilitate reconnection of the device **108** to the central call session, for example by issuing a further central call session notification to the device **108** to prompt the device **108** to rejoin the central call session.

[0067] At block **520**, in response to detected unexpected loss of connectivity, the device **108** may obtain the central call session identifier and initiate a request to rejoin the central call session indicated by the central call session identifier. For example, the device **108** may obtain the central call session identifier from the application cache tracking pending and/or ongoing calls or from a further central call session notification re-issued by the server **104**. The request to rejoin the central call session may be another SIP [INVITE] message or other standard invitation or call initiation according to other signaling protocols.

[0068] At block **525**, in response to the request to rejoin the central call session issued at block **520**, the server **104** may re-add or reconnect the device **108** to the central call session.

[0069] In some examples, if a request to rejoin the central call session is not received by the server **104**, after detecting the unexpected loss of connectivity of the device **108** at block **515**, the server **104** may proceed to block **530** to check for a timeout condition. For example, the timeout condition may include elapsing of a predefined timeout period (e.g., 10 seconds, 30 seconds, etc.) without receiving the request to rejoin the central call session. Further, the timeout condition may check the number of connected participants in the central call session. If the predefined timeout period has elapsed and there is one or fewer connected participants in the central call session, then the server **104** may determine that the timeout condition is met at block **530**.

[0070] If the determination at block **530** is affirmative, that is, the timeout condition is met, then

the server **104** may proceed to block **535** to terminate the central call session. In such examples, the server **104** may provide an indication of termination at the central call session to indicate that the device **108** has been disconnected.

[0071] If the determination at block **530** is negative, that is, the timeout condition is not met, then the server **104** may continue to wait for the request to rejoin the central call session at block **525** and/or return to block **530** to check for the timeout condition.

[0072] Thus, calls established via call sessions hosted centrally at the server **104** may facilitate reconnection of any of the endpoint devices a given call.

[0073] The call sessions hosted centrally at the server **104** may further facilitate other call functionality, such as call merging, to add another endpoint device as a participant to an ongoing call. For example, FIG. **6** depicts a flowchart of an example method **600** of merging a call.

[0074] At block **605**, the server **104** is hosting an ongoing central call session, for example between the devices **108** and **116**. In some examples, the device **108** may be the target device as in the examples described above, while in other examples, the device **108** may be the initiating device of the ongoing call. That is, the method **600** may be applied by the server **104** irrespective of the origination of the ongoing call.

[0075] At block **610**, the server **104** receives a merge request from one of the participant devices of the ongoing central call session, that is, from one of the devices **108** and **116**. The merge request may be similar to a call request and may specify a target account of the merge request. For example, the target may be identified by a target telephone number.

[0076] The server **104** may identify one or more devices associated with the target account. In the present example, the server **104** may identify the device **110** as being associated with the target account of the merge request.

[0077] At block **615**, the server **104** sends a central call session notification to the target device identified at block **310**, in the present example, the device **110**. In particular, the central call session notification includes the call session identifier and may further include an indicator of the device **108**, **116** from which the merge request was received. In some examples, the central call session notification may further include an indicator of the secondary participant(s), that is, the other of the devices **108**, **116** acting as a participant in the ongoing call session.

[0078] The device **110** may receive and process the central call session notification at blocks **620** through **640** similarly as blocks **320** through **640** of the method **300**. That is, at block **620**, the device **110** receives the central call session notification and signals the central call session notification, for example via an alert, including auditory, mechanical, visual alerts, or the like. The alert may provide an indication of the device **108**, **116** from which the merge request was received as the initiator of the call, and in some examples may further indicate that the incoming call includes multiple participants, including specific indicators of the secondary participant(s).

[0079] At block **625**, the device **110** determines whether the call is accepted or rejected based on user input to the alert or expiry of a timeout of the alert.

[0080] If the determination at block **625** is affirmative, the device **110** proceeds to block **630** to send a response to the central call session notification to the server **104**. In particular, the response may include a request to join the central call session, as identified by the central call session identifier, for example via a SIP [INVITE] message or other suitable signaling protocol invitation.

[0081] At block **635**, the server **104** receives the request to join the central call session from the device **110** and in response, adds or connects the device **110** to the central call session identified in the request, effectively merging the call between the merge request initiating device and the device **110** with the call between the devices **108**, **116**.

[0082] If the determination at block **625** is negative, then the device **108** proceeds to block **640** to send a response to the central call session notification to the server **104**. In particular, the response may include a rejection of the central call session notification.

[0083] At block **645**, the server **104** receives the rejection of the central call session from the device

110. In response to the rejection, the server **104** may provide an indication of the rejection at the central call session to indicate that the merge request to the device **110** was rejected. Since the merge request is effectively to add a participant to the ongoing central call session, the server **104** may continue to maintain the central call session between the devices **108**, **116**. Further, since the server **104** may issue the central call session notification to the device **110** independently of the ongoing central call session, the central call session may be maintained while the merge request is being processed. This may improve user experience, since the ongoing central call session allows for continued communications, creating less confusion and fewer dropped calls.

[0084] As described herein, a telecommunications system establishes calls by adding the initiating device to a central call session and sending a prompt (i.e., a central call session notification) to the target device to prompt the target device to request to join the central call session. Accordingly, both endpoint devices of a call may treat the call as an outgoing call, rather than an incoming call, thereby simplifying the initial connection of the call, reconnections in the case of unexpected and/or temporary losses in connectivity, merging calls, and the like. Other applications and advantages of establishing calls using central call sessions will also be apparent to those of skill in the art.

[0085] The scope of the claims should not be limited by the embodiments set forth in the above examples but should be given the broadest interpretation consistent with the description as a whole.

Claims

1. A method comprising: receiving, from an initiating device, a call destined for a target account; establishing, at a server, a central call session for the call and connecting the initiating device to the central call session; sending a central call session notification to a target device associated with the target account, the central call session notification to prompt the target device to initiate a request to join the central call session; receiving, from the target device, a response to the central call session notification; and when the response comprises the request to join the central call session, connecting the target device to the central call session to establish the call.
2. The method of claim 1, further comprising: when the response comprises a rejection, terminating the central call session to end the call.
3. The method of claim 2, further comprising: providing an indication of the rejection to the initiating device via the central call session.
4. The method of claim 1, further comprising: prior to receiving the response, providing a pending indication to the initiating device via the central call session.
5. The method of claim 1, further comprising: detecting an unexpected loss of connectivity of one of the initiating device and the target device; and in response to detecting the unexpected loss of connectivity, when a timeout condition is detected, terminating the central call session.
6. The method of claim 1, further comprising: receiving a merge request for a further target account; sending a further central call session notification to a further target device associated with the further target account, the further central call session notification to prompt the further target device to initiate a further request to join the central call session; receiving, from the further target device, a further response to the further central call session notification; and when the further response comprises the further request to join the central call session, connecting the further target device to the central call session to merge further target device to the call.
7. A method comprising: receiving, at a target device, a central call session notification from a server, the central call session notification indicating a call; signaling the central call session notification to prompt a response to the central call session notification; and when the response comprises an acceptance of the call, sending a request to join a central call session identified in the central call session notification to the server to accept the call.
8. The method of claim 7, further comprising: when the response comprises a rejection of the call,

sending an indication of the rejection to the server to reject the call.

9. The method of claim 7, wherein the central call session notification indicates the call as an incoming call from an initiating device.

10. The method of claim 7, further comprising: after joining the central call session, detecting an unexpected loss of connectivity to the central call session; and sending a further request to rejoin the central call session to the server to reconnect to the call.

11. A server comprising: a memory and a communications interface; a processor interconnected with the memory and the communications interface, the processor configured to: receive, from an initiating device, a call destined for a target account; establish a central call session for the call and connecting the initiating device to the central call session; send a central call session notification to a target device associated with the target account, the central call session notification to prompt the target device to initiate a request to join the central call session; receive, from the target device, a response to the central call session notification; and when the response comprises the request to join the central call session, connect the target device to the central call session to establish the call.

12. The server of claim 11, wherein the processor is further configured to: when the response comprises a rejection, terminate the central call session to end the call.

13. The server of claim 12, wherein the processor is further configured to: provide an indication of the rejection to the initiating device via the central call session.

14. The server of claim 11, wherein the processor is further configured to: prior to receiving the response, provide a pending indication to the initiating device via the central call session.

15. The server of claim 11, wherein the processor is further configured to: detect an unexpected loss of connectivity of one of the initiating device and the target device; and in response to detecting the unexpected loss of connectivity, when a timeout condition is detected, terminate the central call session.

16. The server of claim 11, wherein the processor is further configured to: receive a merge request for a further target account; send a further central call session notification to a further target device associated with the further target account, the further central call session notification to prompt the further target device to initiate a further request to join the central call session; receive, from the further target device, a further response to the further central call session notification; and when the further response comprises the further request to join the central call session, connect the further target device to the central call session to merge further target device to the call.

17. A computing device comprising: a memory and a communications interface; and a processor interconnected with the memory and the communications interface, the processor configured to: receive, at a target device, a central call session notification from a server, the central call session notification indicating a call; signal the central call session notification to prompt a response to the central call session notification; and when the response comprises an acceptance of the call, send a request to join the central call session to the server to accept the call.

18. The computing device of claim 17, wherein the processor is further configured to: when the response comprises a rejection of the call, send an indication of the rejection to the server to reject the call.

19. The computing device of claim 17, wherein the central call session notification indicates the call as an incoming call from an initiating device.

20. The computing device of claim 17, wherein the processor is further configured to: after joining the central call session, detect an unexpected loss of connectivity to the central call session; and send a further request to rejoin the central call session to the server to reconnect to the call.
